Lecture 4

Continuity

A function is said to be continuous at a point $x = a$ if:

(i) $f(a)$ is defined.
(ii) $\lim_{x\to a} f(x)$ exists (two-sided limit exists).
(iii) $f(a) = \lim_{x\to a} f(x)$.

- $f(a) = LHL = RHL \Rightarrow f(x)$ is continuous at a and the limit also exists.
- $f(a) \neq LHL = RHL, \Rightarrow$ the limit exists at a, but $f(x)$ is not continuous at a.

Example: $f(x) = \begin{cases} x^2 + 1, & ; x > 0 \\ 1, & ; x = 0 \\ 1 + x, & ; x < 0 \end{cases}$

Is $f(x)$ continuous at $x = 0$?

Answer: Here, at $x = 0$, $f(0) = 1$.

So, $f(0)$ is defined.

Now, LHL: $\lim_{x\to 0^-} f(x) = \lim_{x\to 0^-} (1 + x) = 1 + 0 = 1$

RHL: $\lim_{x\to 0^+} f(x) = \lim_{x\to 0^+} (x^2 + 1) = 0^2 + 1 = 1$

So, the limit exists at x=0 and $\lim_{x\to 0} f(x) = 1$.

We can see that, $f(0) = \lim_{x\to 0} f(x) = 1$.

Hence, the function $f(x)$ is continuous at $x = 0$.

Example:

$$f(x) = \begin{cases} e^{-\frac{|x|}{2}} & \text{if } -1 < x < 0 \\ x^2 & \text{if } 0 \leq x < 2 \end{cases}$$

Test continuity at $x = 0$.

Answer: Here, at $x = 0$, $f(0) = 0^2 = 0$.

So, $f(0)$ is defined.

Now, LHL: $\lim_{x\to 0^-} f(x) = \lim_{x\to 0^-} e^{-\frac{|x|}{2}}$

$= \lim_{x\to 0^-} e^{-\frac{(-x)}{2}}$ [As the interval is $-1 < x < 0$, so, $x < 0$ here]

$$= \lim_{x\to 0^-} e^{\frac{x}{2}} = e^0 = 1$$

$$\text{RHL:} \quad \lim_{x\to 0^+} f(x) = \lim_{x\to 0^+} x^2 = 0^2 = 0$$

Since RHL $\neq$ LHL, the limit does not exist and the function is not continuous at $x = 0$.

Example:

$$f(x) = \begin{cases} \dfrac{|x-3|}{x-3} & ; x \neq 3 \\ 0 & ; x = 3 \end{cases}$$

Test continuity at $x = 3$.

Answer: Here, $|x-3| = \begin{cases} x-3 & ; x-3 > 0 \\ -(x-3) & ; x-3 < 0 \end{cases} = \begin{cases} x-3 & ; x > 3 \\ -(x-3) & ; x < 3 \end{cases}$

$$\text{Now, } f(x) = \begin{cases} \frac{-(x-3)}{x-3} & ; x < 3 \\ 0 & ; x = 3 \\ \frac{x-3}{x-3} & ; x > 3 \end{cases}$$

$$= \begin{cases} -1 & ; x < 3 \\ 0 & ; x = 3 \\ 1 & ; x > 3 \end{cases}$$

So, $f(3) = 0$.

$$\text{LHL:} \quad \lim_{x\to 3^-} f(x) = \lim_{x\to 3^-} (-1) = -1$$

$$\text{RHL:} \quad \lim_{x\to 3^+} f(x) = \lim_{x\to 3^+} (1) = 1$$

As LHL $\neq$ RHL, the limit does not exist and the function is not continuous at $x = 3$.

<u>The Squeezing Theorem:</u>

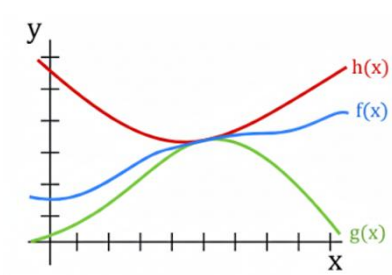
Let f, g and h be functions satisfying $g(x) \leq f(x) \leq h(x)$ for all x in some open interval containing the number a, with possible exception that the inequalities need not hold at a.

If g and h have the same limits as x approaches a, say

$$\lim_{x\to a} g(x) = \lim_{x\to a} h(x) = L,$$

Then f also has limit ax x approaches a, that is,

$$\lim_{x\to a} f(x) = L.$$



The squeezing theorem is especially useful for finding limits of trigonometric functions.

Example: $\lim_{x\to 0} x\sin\left(\frac{1}{x}\right) = ?$

Solution: Since,

$$-1 \le \sin\left(\frac{1}{x}\right) \le 1$$

$$\text{Or, } -|x| \le x\sin\left(\frac{1}{x}\right) \le |x| \qquad [\text{when } x \ne 0]$$

Let , $g(x) = -|x|$

$f(x) = x\sin\left(\frac{1}{x}\right)$

$h(x) = |x|$.

Now, $\lim_{x\to 0} g(x) = \lim_{x\to 0}(-|x|) = 0$

$\lim_{x\to 0} h(x) = \lim_{x\to 0}|x| = 0.$

Hence, according to squeeze theorem we can say that, $\lim_{x\to 0} x\sin\left(\frac{1}{x}\right) = 0$.

H.W.

1. $\lim_{x\to 0} x\cos\left(\frac{1}{x}\right) = ?$ Answer: 0
2. $\lim_{x\to 0} x^2\sin\left(\frac{1}{x}\right) = ?$ Answer: 0
3. $\lim_{x\to +\infty} \frac{\sin x}{x} = ?$ Answer: 0

Example: $f(x) = \begin{cases} x\sin\left(\frac{1}{x}\right), & x \ne 0 \\ 0, & x = 0 \end{cases}$

Test continuity at $x = 0$.

Solution: Here, $f(0) = 0$.

So, $f(0)$ is defined.

$$\text{Since,} \qquad -1 \le \sin\left(\frac{1}{x}\right) \le 1$$

$$\text{Or, } -|x| \le x\sin\left(\frac{1}{x}\right) \le |x| \qquad [\text{when } x \ne 0]$$

Let , $g(x) = -|x|$

$f(x) = x\sin\left(\frac{1}{x}\right)$

$h(x) = |x|$

Now, $\lim_{x\to 0} g(x) = \lim_{x\to 0}(-|x|) = 0$

$$\lim_{x\to 0} h(x) = \lim_{x\to 0} |x| = 0.$$

Hence, according to squeeze theorem we can say that, $\lim_{x\to 0} x\sin\left(\frac{1}{x}\right) = 0$.

Here , $f(0) = \lim_{x\to 0} x\sin\left(\frac{1}{x}\right) = 0$.

so, the function $f(x)$ is continuous at $x = 0$.

Example: $f(x) = \begin{cases} (x-a)\sin\left(\frac{1}{x-a}\right), & x \neq a \\ 0, & x = 0 \end{cases}$

Test continuity at $x = a$.

Solution: Here, $f(a) = 0$.

So, $f(a)$ is defined.

Let, $h(x) = x - a$, then, as $x \to a, h \to 0$.

So we get, $(x-a)\sin\left(\frac{1}{x-a}\right) = h\sin(\frac{1}{h})$

Since, $\quad -1 \leq \sin\frac{1}{h} \leq 1$

Or, $-|h| \leq h\sin\left(\frac{1}{h}\right) \leq |h| \qquad$ [when $h \neq 0$]

Now, $\quad \lim_{h\to 0}(-|h|) = 0$

$$\lim_{h\to 0}|h| = 0.$$

Hence, according to squeeze theorem we can say that, $\lim_{h\to 0} h\sin\left(\frac{1}{h}\right) = \lim_{x\to a}(x-a)\sin\left(\frac{1}{x-a}\right) = 0$.

So, $f(a) = \lim_{x\to a} f(x) = 0$.

Hence, the function $f(x)$ is continuous at $x = a$.

H.W. $f(x) = \begin{cases} 1, & x < 0 \\ 1 + \sin x, & 0 \leq x < \frac{\pi}{2} \\ 2 + \left(x - \frac{\pi}{2}\right)^2, & x \geq \frac{\pi}{2} \end{cases}$

Test continuity at $x = 0$ and $x = \frac{\pi}{2}$.

Some Properties of Continuous Functions:

1.5.3 THEOREM *If the functions f and g are continuous at c, then*

(a) $f + g$ *is continuous at* c.

(b) $f - g$ *is continuous at* c.

(c) fg *is continuous at* c.

(d) f/g *is continuous at* c *if* $g(c) \neq 0$ *and has a discontinuity at* c *if* $g(c) = 0$.

Continuity of Polynomials and Rational Functions:

> **1.5.4 THEOREM**
>
> (*a*) *A polynomial is continuous everywhere.*
>
> (*b*) *A rational function is continuous at every point where the denominator is nonzero, and has discontinuities at the points where the denominator is zero.*

Right Continuous at a:

$\lim_{x \to a^+} f(x) = f(a)$

Left Continuous at a:

$\lim_{x \to a^-} f(x) = f(a)$

Continuous from both right and left at a:

$\lim_{x \to a^+} f(x) = f(a)$ and $\lim_{x \to a^-} f(x) = f(a)$.

So, $\lim_{x \to a^+} f(x) = \lim_{x \to a^-} f(x) = f(a)$.

So, we can say that, $f(x)$ is continuous at $x = a$.

Continuity of a Function on an Interval:

A function $f(x)$ is said to be continuous on a closed interval $[a, b]$ if

(i) $f(x)$ is continuous on (a, b).

(A function $f(x)$ is continuous at the interval (a, b) if $f(x)$ is continuous at each point of the interval.)

(ii) $f(x)$ is continuous from the right (right continuous) at $x = a$, i.e.,

$$\lim_{x \to a^+} f(x) = f(a)$$

(iii) $f(x)$ is continuous from the left (left continuous) at $x = b$, i.e.,

$$\lim_{x \to b^-} f(x) = f(b).$$

Example: $f(x) = \sqrt{9 - x^2}, \quad -3 \le x \le 3$

Show that the function $f(x)$ is continuous on the interval $[-3,3]$.

Solution:

(i) $f(x)$ is continuous on $(-3,3)$.

(ii) $\lim_{x \to -3^+} f(x) = \lim_{x \to -3^+} \sqrt{9 - x^2} = \sqrt{9 - (-3)^2} = 0$

and $f(-3) = \sqrt{9-(-3)^2} = 0$

So, we get, $\lim_{x\to -3^+} f(x) = f(-3) = 0$.

Hence, $f(x)$ is continuous from the right (right continuous) at $x = -3$.

(iii) $\lim_{x\to 3^-} f(x) = \lim_{x\to 3^-} \sqrt{9-x^2} = \sqrt{9-(3)^2} = 0$

and $f(3) = \sqrt{9-(3)^2} = 0$

So, we get, $\lim_{x\to 3^-} f(x) = f(3) = 0$.

Hence, $f(x)$ is continuous from the left (left continuous) at $x = 3$.

So, $f(x)$ is continuous on the interval $-3 \le x \le 3$.

H.W.

1. $f(x) = x^2, \quad 1 \le x \le 5$. Show that the function $f(x)$ is continuous on the interval [1,5].
2. $f(x) = \sqrt{4x-3}$, Show that the function $f(x)$ is continuous on the interval $[1, 3]$.

Example: Prove that the function $f(x)$ is right continuous at $x = 0$ but not continuous at $x = 0$.

Given, $f(x) = \begin{cases} \frac{|x|}{x}, & x \neq 0 \\ 1, & x = 0 \end{cases}$

Solution: We can write,

$$f(x) = \begin{cases} \frac{-x}{x}, & x < 0 \\ 1, & x = 0 \\ \frac{x}{x}, & x > 0 \end{cases}$$

$$= \begin{cases} -1, & x < 0 \\ 1, & x = 0 \\ 1, & x > 0 \end{cases}$$

Here, $f(0) = 1$.

So, $f(0)$ is defined.

RHL = $\lim_{x\to 0^+} f(x) = \lim_{x\to 0^+} (1) = 1$

LHL = $\lim_{x\to 0^-} f(x) = \lim_{x\to 0^-} (-1) = -1$

Here, $f(0) = \lim_{x\to 0^+} f(x) = 1$

So, the function is right continuous at $x = 0$, but since LHL $\neq$ RHL, the function is not continuous at $x = 0$.

Example: Find the value of the constant k, if possible, that will make the function $f(x)$continuous everywhere.

$$f(x) = \begin{cases} kx^2, & x \leq 2 \\ 2x + k, & x > 2 \end{cases}$$

Solution: Here, f is continuous for $x < 2$, and for $x > 2$.

For continuity at $x = 2$, we require, LHL = RHL = $f(2)$.

Here, $f(2) = k(2)^2 = 4k$

LHL $= \lim_{x \to 2^-} f(x) = \lim_{x \to 2^-} kx^2 = k(2)^2 = 4k$

RHL $= \lim_{x \to 2^+} f(x) = \lim_{x \to 2^+} (2x + k) = 2(2) + k = 4 + k$

<u>For the value of k:</u>

$f(2) = \text{RHL}$	LHL = RHL
Or, $4k = 4 + k$	Or, $4k = 4 + k$
Or, $4k - k = 4$	Or, $4k - k = 4$
Or, $3k = 4$	Or, $3k = 4$
So, $k = \frac{4}{3}$	So, $k = \frac{4}{3}$

So, the value $k = \frac{4}{3}$, $w$ill make the function $f(x)$ everywhere continuous. .

H.W. Exercise 1.5 (29-30)

Example: $f(x) = \begin{cases} \frac{sin^2 ax}{x^2}, & x \neq 0 \\ 1, & x = 0 \end{cases}$

Prove that if $a = 1$, then the function is continuous.

Solution:

For continuity at $x = 0$, we require, L.H.L = R.H.L =$f(0)$.

At $x = 0, f(x) = f(0) = 1$.

A function may be discontinuous at the points where its definition changes from one interval to another. Therefore, to ensure that a function is continuous, continuity must be checked specifically at those points where the intervals change. The question will not explicitly mention the points at which you need to check continuity, you have to identify them yourself.

$$\text{L.H.L} = \lim_{x\to 0^-} \frac{sin^2 ax}{x^2}$$
$$= \lim_{x\to 0^-} \left(\frac{sinax}{x}\right)^2$$
$$= \lim_{x\to 0^-} \left(\frac{sinax}{ax} \times a\right)^2$$
$$= \lim_{x\to 0^-} \left(\frac{sinax}{ax}\right)^2 \times a^2$$
$$= \left(\lim_{x\to 0^-} \frac{sinax}{ax}\right)^2 \times a^2$$
$$= (1)^2 \times a^2 = a^2$$

$$\text{R.H.L} = \lim_{x\to 0^+} \frac{sin^2 ax}{x^2}$$
$$= \lim_{x\to 0^+} \left(\frac{sinax}{x}\right)^2$$
$$= \lim_{x\to 0^+} \left(\frac{sinax}{ax} \times a\right)^2$$
$$= \lim_{x\to 0^+} \left(\frac{sinax}{ax}\right)^2 \times a^2$$
$$= \left(\lim_{x\to 0^+} \frac{sinax}{ax}\right)^2 \times a^2$$
$$= (1)^2 \times a^2 = a^2$$

$$\boxed{\lim_{x\to 0} \frac{sinax}{ax} = 1}$$

Now, if $a = 1$, L.H.L $= a^2 = (1)^2 = 1$.

And R.H.L $= a^2 = (1)^2 = 1$.

So, $f(0) = \text{LHL} = \text{RHL} = 1$.

Hence, if $a = 1$, $f(x)$ is continuous. (Proved)

Example: Find values of x, if any, at which f is not continuous.

$$f(x) = \begin{cases} 2x + 3, & x \le 4 \\ 7 + \dfrac{16}{x}, & x > 4 \end{cases}$$

Solution: Here, $f(x) = 2x + 3$ is continuous on $x < 4$ and $f(x) = 7 + \frac{16}{x}$ is continuous on $4 < x$.

For continuity at $x = 4$, L.H.L = R.H.L $= f(4)$.

At $x = 4$, $f(x) = 2 \times 4 + 3 = 8 + 3 = 11$.

$$\text{L.H.L} = \lim_{x\to 4^-} (2x + 3)$$
$$= 2 \times 4 + 3$$
$$= 11$$

$$\text{R.H.L} = \lim_{x\to 4^+} (7 + \tfrac{16}{x})$$
$$= 7 + \frac{16}{4}$$
$$= 11$$

So, no such values can be found. $f(x)$ is a continuous function in the real number.

H.W. Exercise 1.5 (21-22)